

THE CHAMELEON

Adaptive AI-Powered Honeypot & Cyber Deception System

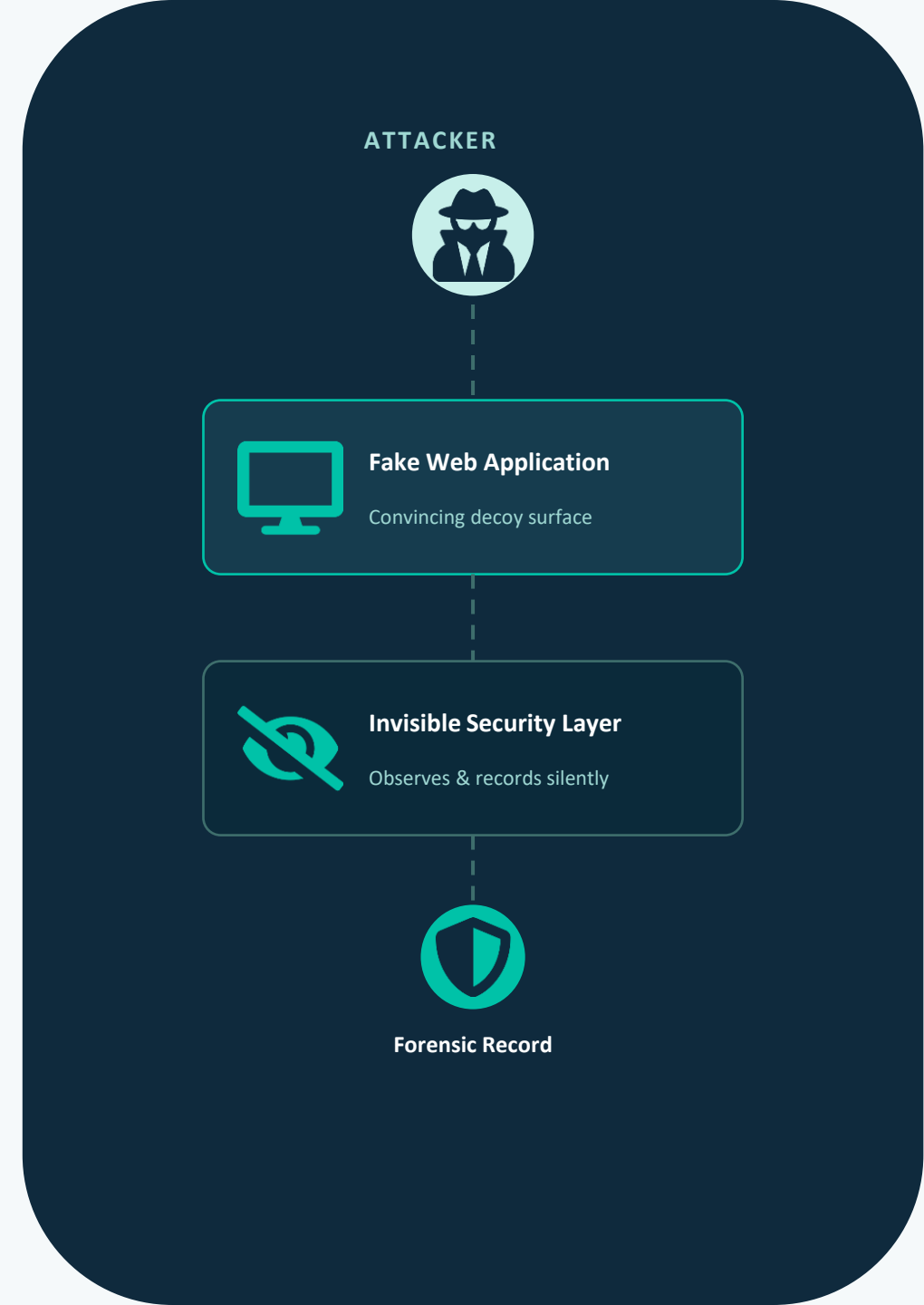
*"Most security tools block attackers.
Ours lies to them."*

TEAM NAME

[Binary Builders]

TEAM MEMBERS

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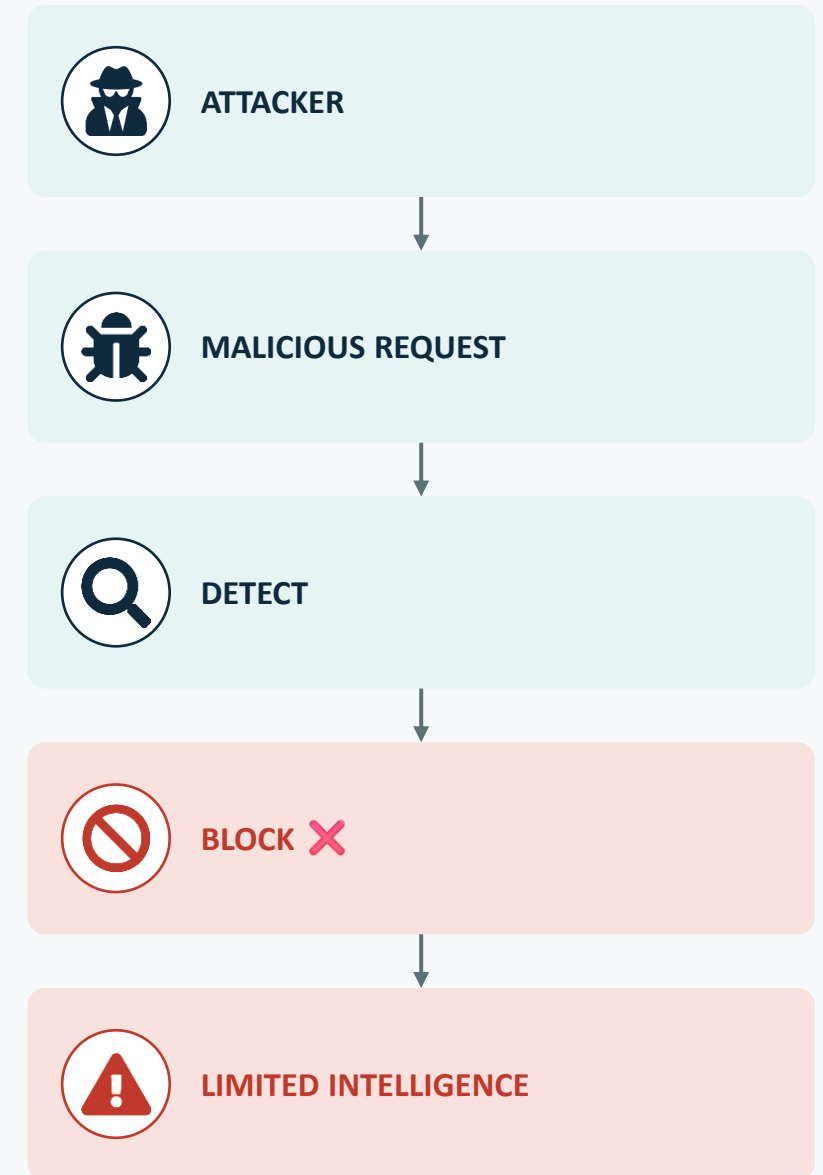


The Problem

Traditional security systems mainly detect and block malicious requests.

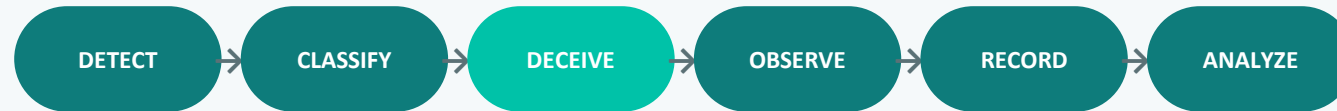
- Blocking attackers can limit visibility into their behavior.
- Security teams may not know what the attacker intended to do.
- Attackers can repeatedly change their payloads and techniques.
- Raw security logs can be difficult to analyze and correlate.
- Security teams need better attack intelligence and forensic visibility.

“Can we detect an attacker without immediately revealing that they have been detected?”

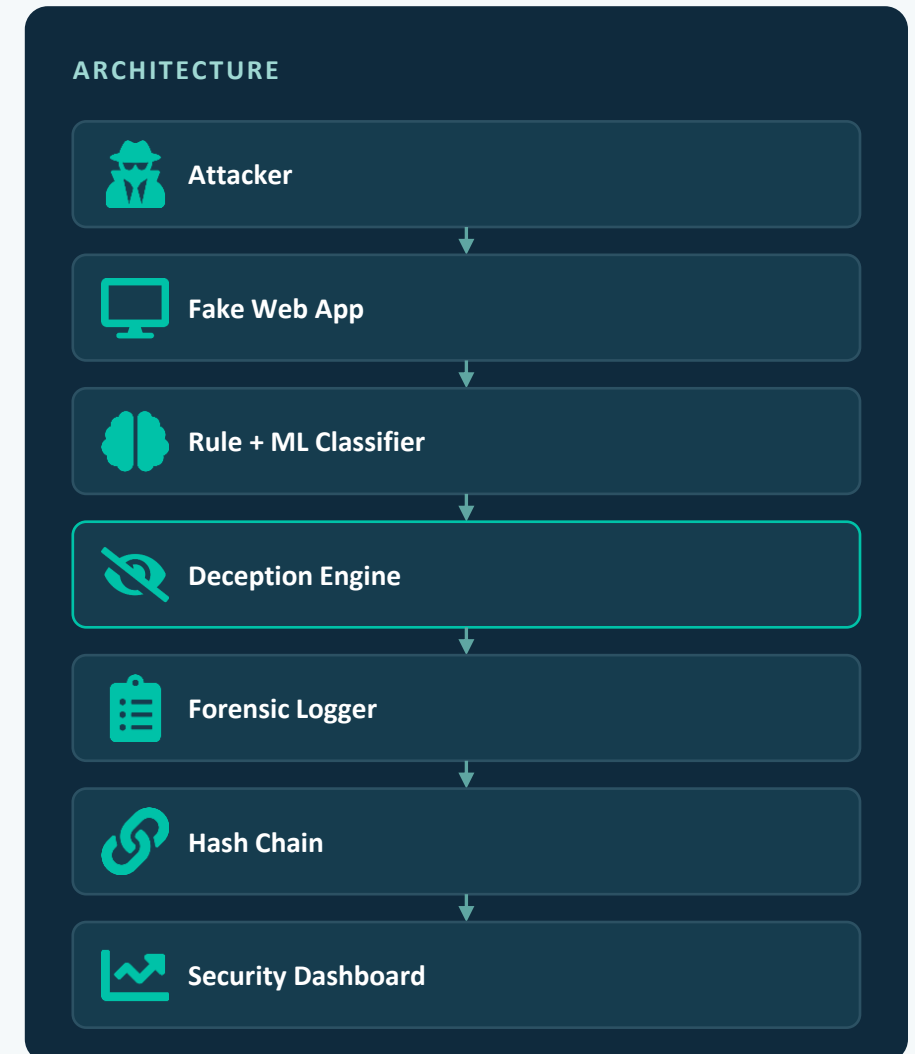


Our Solution

THE CHAMELEON is an adaptive web honeypot that safely deceives attackers while collecting information about their behavior.



- 1 A fake login/search/admin interface acts as the honeypot.
- 2 SQL Injection and XSS inputs are detected using rules + ML.
- 3 Malicious users receive safe, convincing fake responses.
- 4 Their interactions are recorded in real time.
- 5 Events are linked using SHA-256 hashing so tampering becomes detectable.
- 6 Security analysts view the complete attack through a dashboard.



Technology Stack

Each layer of the data flow, mapped to the technology that powers it.



FRONTEND

React / TypeScript / Tailwind CSS



BACKEND

Python / FastAPI



AI / ML

Scikit-learn — SQLi + XSS classification
Rule-based detection + ML



DECEPTION

Python adaptive response engine
Optional LLM for fabricated data



FORENSICS

SHA-256 hash chain
SQLite / JSON



DASHBOARD

React — Chart.js / Recharts
WebSocket / polling



REPORTING

ReportLab / WeasyPrint

What Makes THE CHAMELEON Different?

1. Adaptive Deception



SQLi Login → Fake Successful Login
Data Extraction → Fake Database Records
XSS Attempt → Safe Fake Acceptance
Repeated Attempts → Progressive Delay

2. Hybrid AI Detection



Rules provide fast detection while ML helps classify variations.

3. Attacker Behavior Visibility



Instead of immediately terminating the interaction, the system safely observes the attacker's actions.

4. Tamper-Evident Forensics



Attack events are linked using SHA-256 hashes so modifications to historical records can be detected.

5. Real-Time Security Dashboard



Analysts see the full chain: Attack → Deception → Behavior → Evidence.

"Instead of simply blocking an attacker, THE CHAMELEON turns the attacker's interaction into a controlled source of threat intelligence."

Impact & Real-World Applications



Cyber Defense

Improve visibility into suspicious web activity.



Threat Intelligence

Understand attack patterns and attacker behavior.



SOC Training

A safe environment for analysts to practice investigation.



Cybersecurity Education

Demonstrate SQLi, XSS, deception, and forensic analysis.



Incident Response

Generate structured incident information for investigation.

TRADITIONAL SECURITY

Detect → Block

THE CHAMELEON

Detect → Deceive → Observe → Preserve → Learn

“Don't just block the attack. Make the attack reveal itself.”